The Planck-Scale Structure of the Conversion Factors

Why $G = (\ell_P^2 \times c^3)/\hbar$ justifies the form of the factors from Document 012

T0 Theory: From Dimensionless to SI

Contents

Abstract

This document explains why the conversion factors in Document 012 have the exact form they do. The mathematical relation $G = \frac{\ell_P^2 \times c^3}{\hbar}$ is not a new calculation method (it is a rearrangement of the known Planck length definition), but it reveals the *fundamental structure* underlying the conversion factors.

Core Message: The factors C_{dim}, C_{conv}, and K_{frak} in Document 012 are not arbitrary but follow from the Planck-scale structure of G. The formula also serves as a consistency check: if all factors are correct, $G_{\text{SI}} = \frac{\ell_P^2 \times c^3}{\hbar}$ must be satisfied.

For the complete technical derivation of all conversion factors, see Document 012.

.1 The Problem: Conversion from T0 to SI

Recap: The T0 Formula for G

From Document 012 it is known:

$$G_{\text{SI}} = \frac{\xi^2}{4m_e} \times C_{\text{dim}} \times C_{\text{conv}} \times K_{\text{frak}} \tag{1}$$

With the factors:

- $\frac{\xi^2}{4m_e} \approx 8.7 \times 10^{-9}$ MeV^{-1} (from T0 geometry)
- $C_{\text{dim}} \approx 3.5 \times 10^{-2}$ (dimension correction)
- $C_{\text{conv}} \approx 7.8 \times 10^{-3}$ m^{3}kg^{-1}s^{-2}·MeV (SI conversion)
- $K_{\text{frak}} = 0.986$ (fractal correction)

The Question

Why do these factors have exactly this form?

Specifically:

- Why does c^3 appear? (in C_{conv})
- Why $\hbar$ in the denominator?
- Why a length scale squared?
- What is the fundamental structure?

.2 The Planck Length as the Starting Point

Standard Definition (since Max Planck, 1899)

The Planck length is defined as:

$$\ell_P = \sqrt{\frac{\hbar G}{c^3}} \tag{2}$$

Standard Interpretation:

- G is a fundamental constant (measured)
- ℓ_P is calculated from it
- $\ell_P \approx 1.616 \times 10^{-35}$ m
- Quantum gravity scale

Mathematical Rearrangement

From $\ell_P = \sqrt{\frac{\hbar G}{c^3}}$ it follows by rearrangement:

$$\ell_P^2 = \frac{\hbar G}{c^3} \tag{3}$$

$$\ell_P^2 \times c^3 = \hbar G \tag{4}$$

$$G = \frac{\ell_P^2 \times c^3}{\hbar} \tag{5}$$

This is the fundamental structure!

.3 The Structure of the Conversion Factors

What Does the Planck Formula Show?

[Fundamental Structure]

$$\boxed{G = \frac{\ell_P^2 \times c^3}{\hbar}} \tag{6}$$

Dimensional Analysis:

$$[G] = \frac{[\ell_P^2] \times [c^3]}{[\hbar]} \tag{7}$$

$$= \frac{[\mathrm{m}^2] \times [\mathrm{m}^3/\mathrm{s}^3]}{[\mathrm{J} \cdot \mathrm{s}]} \tag{8}$$

$$= \frac{[\mathrm{m}^5/\mathrm{s}^3]}{[\mathrm{kg} \cdot \mathrm{m}^2/\mathrm{s}^2 \cdot \mathrm{s}]} \tag{9}$$

$$= \frac{[\mathrm{m}^5/\mathrm{s}^3]}{[\mathrm{kg} \cdot \mathrm{m}^2/\mathrm{s}]} \tag{10}$$

$$= \frac{[\mathrm{m}^3]}{[\mathrm{kg} \cdot \mathrm{s}^2]} \tag{11}$$

Exactly [G] = $\mathrm{m}^3/(\mathrm{kg} \cdot \mathrm{s}^2)$! ✓

Connection to T0 Factors

In T0, one starts with G_{nat} in dimension $[E^{-2}]$ (Energy^{-2}).
Conversion $[E^{-2}] \rightarrow$ [m³/(kg·s²)] must have:

$$[E^{-2}] \times \text{Factor} = [\text{m}^3/(\text{kg} \cdot \text{s}^2)] \tag{12}$$

The factor must have the structure:

$$\text{Factor} = \frac{[\text{Length}^3]}{[\text{Energy}]} \tag{13}$$

From the Planck formula:

$$G = \frac{\ell_P^2 \times c^3}{\hbar} \quad \Rightarrow \quad \text{Structure: } \frac{[\text{Length}^2] \times [\text{Velocity}^3]}{[\text{Action}]} \tag{14}$$

With $[\hbar] = [\text{Energy} \times \text{Time}]$ and $[c] = [\text{Length}/\text{Time}]$:

$$\frac{[\ell_P^2 \times c^3]}{[\hbar]} = \frac{[\text{Length}^2] \times [\text{Length}^3/\text{Time}^3]}{[\text{Energy} \times \text{Time}]} \tag{15}$$

$$= \frac{[\text{Length}^5/\text{Time}^3]}{[\text{Energy} \times \text{Time}]} \tag{16}$$

$$= \frac{[\text{Length}^5]}{[\text{Energy} \times \text{Time}^4]} \tag{17}$$

This justifies why:

- c^3 in the numerator (Length³/Time³)
- $\hbar$ in the denominator (Energy × Time)
- Length² (from ℓ_P^2)
- The combination yields [m³/(kg·s²)]

.4 Justification of the Factors in Document 012

The Dimension Correction Factor C_{dim}

From Document 012:

$$C_{\text{dim}} = \frac{1}{E_{\text{char}}} \approx 3.5 \times 10^{-2} \quad [\text{MeV}^{-1}] \tag{18}$$

With $E_{\text{char}} = 28.4$ MeV (7-step derivation in Doc. 012).
Why this factor?
The T0 formula $G = \frac{\xi^2}{4m_e}$ initially yields dimension $[E^{-1}]$.
But G needs $[E^{-2}]$ in natural units!
$\Rightarrow$ Factor $[E^{-1}]$ needed: $C_{\text{dim}} = 1/E_{\text{char}}$
Connection to the Planck structure:
The energy scale E_{char} is not arbitrary but emerges from the same geometry as ℓ_P. It is the characteristic scale where T0 geometry connects to the Planck scale.

The SI Conversion Factor C_{conv}

From Document 012:

$$C_{\text{conv}} \approx 7.8 \times 10^{-3} \quad [\text{m}^3\text{kg}^{-1}\text{s}^{-2} \cdot \text{MeV}] \tag{19}$$

Structure of this factor:

$$C_{\text{conv}} \sim \frac{c^3}{\hbar} \quad \text{(in appropriate units)} \tag{20}$$

$$= \frac{(2.998 \times 10^8)^3}{1.055 \times 10^{-34}} \quad \text{(order of magnitude)} \tag{21}$$

Why exactly this combination?

The Planck formula $G = \frac{\ell_P^2 \times c^3}{\hbar}$ shows:

- c^3 converts time scale to space scale (dimension: m^3/s^3)
- $\hbar$ connects energy with frequency (dimension: J·s)
- Combination $c^3/\hbar$ has dimension $[m^3/(kg \cdot s^2)]/[\text{Energy}]$

Exactly what C_{conv} provides!

Numerical Verification

Consistency Check

From T0 (Document 012):

$$G_{\text{nat}} = \frac{\xi^2}{4m_e} \times C_{\text{dim}} \approx 3.1 \times 10^{-10} \quad [E^{-2}] \tag{22}$$

$$G_{\text{SI}} = G_{\text{nat}} \times C_{\text{conv}} \times K_{\text{frak}} \tag{23}$$

$$\approx 3.1 \times 10^{-10} \times 7.8 \times 10^{-3} \times 0.986 \times 10^1 \tag{24}$$

$$\approx 6.67 \times 10^{-11} \quad [\text{m}^3/(\text{kg} \cdot \text{s}^2)] \tag{25}$$

From Planck Formula (Verification):

$$\ell_P = 1.616 \times 10^{-35} \text{ m} \tag{26}$$

$$c = 2.998 \times 10^8 \text{ m/s} \tag{27}$$

$$\hbar = 1.055 \times 10^{-34} \text{ J·s} \tag{28}$$

$$G_{\text{check}} = \frac{\ell_P^2 \times c^3}{\hbar} \tag{29}$$

$$= \frac{(1.616 \times 10^{-35})^2 \times (2.998 \times 10^8)^3}{1.055 \times 10^{-34}} \tag{30}$$

$$= \frac{2.611 \times 10^{-70} \times 2.694 \times 10^{25}}{1.055 \times 10^{-34}} \tag{31}$$

$$= 6.67 \times 10^{-11} \quad [\text{m}^3/(\text{kg} \cdot \text{s}^2)] \tag{32}$$

Perfect agreement! ✓

This shows: The factors in Doc. 012 have exactly the right structure.

.5 The Role of the Planck Formula in T0

Not Circular in T0

Why is the formula not circular?
Standard Physics (circular):

1. Measure G
2. Calculate $\ell_P = \sqrt{\hbar G/c^3}$
3. Calculate $G = \ell_P^2 c^3/\hbar$
   ⇒ Get G back (useless!)

T0 Physics (not circular):

1. Determine ξ from experiment (via α, E_0)
2. Calculate G_{SI} from ξ (with factors)
3. Calculate $\ell_P = \sqrt{\hbar G_{\text{SI}}/c^3}$
4. Check: $G_{\text{SI}} = \ell_P^2 c^3/\hbar$
   ⇒ Consistency check! ✓

Three Uses of the Planck Formula

1. **Justification:** Shows why factors have the form $c^3/\hbar$ etc.
2. **Verification:** Consistency check for calculated G
3. **Structural Insight:** G emerges at the Planck scale

.6 Practical Application: Python Implementation

Code Structure (from calc_De.py)

The T0 calculation script shows exactly this logic:

```
# Main calculation (from ξ)
G_t0_dimensionless = (xi**2) / (4 * m_char)
conversion_factor_nat = 3.521e-2  # C_dim
G_nat = G_t0_dimensionless * conversion_factor_nat

SI_conversion_factor = 2.843e-5   # C_conv × K_frak
G_SI = G_nat * SI_conversion_factor

# Planck formula as verification
planck_conversion_factor = (l_P**2 * c**3) / hbar

# Check: Both should agree!
assert abs(G_SI - planck_conversion_factor) < 1e-13
```

What the Code Shows

- **Lines 1-2:** T0 formula $\xi^2/(4m)$
- **Line 3:** Dimension correction C_{dim} (corresponds to $1/E_{\text{char}}$)

- **Line 5:** SI conversion $C_{\text{conv}} \times K_{\text{frak}}$ (corresponds to $c^3/\hbar$ structure)
- **Line 8:** Planck formula for verification
- **Line 11:** Both paths must agree!

.7 Comparison with Electrodynamics

Analogy: Speed of Light

In electrodynamics:

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} \tag{33}$$

Interpretation:

- c emerges from electromagnetic vacuum structure
- μ_0, ε_0 describe vacuum properties
- Formula shows structure, not calculation

Analogy: Gravitational Constant

In T0:

$$G = \frac{\ell_P^2 \times c^3}{\hbar} \tag{34}$$

Interpretation:

- G emerges from spacetime geometry (T0)
- ℓ_P, c, $\hbar$ describe geometry properties
- Formula shows structure, justifies conversion factors

Parallelism

Aspect	Electrodynamics	Gravitation
Constant	c	G
Formula	$c = 1/\sqrt{\mu_0\varepsilon_0}$	$G = \ell_P^2 c^3/\hbar$
Emerges from	EM vacuum	Spacetime geometry
Justifies	μ_0, ε_0 structure	C_{conv} structure

Table 1: Parallel Structures

.8 Summary

The Central Message

[Structure Justification] **The Planck formula $G = \frac{\ell_P^2 \times c^3}{\hbar}$ is essential for T0 because it:**

1. **Justifies** why the conversion factors in Doc. 012 have exactly the form:
   - $C_{\text{dim}} \sim 1/E$ (energy scale)
   - $C_{\text{conv}} \sim c^3/\hbar$ (Planck structure)

2. **Serves as a consistency check:**
   - Calculate G from ξ with factors
   - Calculate ℓ_P from G
   - Check: $G = \ell_P^2 c^3/\hbar$ ✓
3. **Shows the geometric structure:**
   - G emerges at Planck scale ℓ_P
   - Connection quantum mechanics ($\hbar$) $\leftrightarrow$ relativity (c)
   - Fundamental role of geometry

It is not a new calculation method (would be circular), but it is the justification for the factor structure!

What is New?

Mathematically NOT new:
- The formula $G = \ell_P^2 c^3/\hbar$ (rearrangement of ℓ_P definition since 1899)
- The Planck units (Max Planck, 1899)

New in T0:
- The formula *justifies* the conversion factors
- It serves as *verification* (not circular, since G comes from ξ)
- It shows that G emerges at the Planck scale
- ℓ_P is not fundamental but follows from G (which follows from ξ)

Connection to Document 012

Document 012 shows: HOW to calculate G from ξ (all steps)
This document (127) shows: WHY the factors have this structure
Together: Complete picture of G in T0

Practical Significance

For calculations:
- Use the T0 path: $\xi \rightarrow G$ (Doc. 012)
- Planck formula as a check
- Both must agree

For understanding:
- Planck formula shows structure
- Justifies why $c^3/\hbar$ appears
- Shows geometric origin

For philosophy:
- G is not fundamental
- G emerges at the Planck scale
- Everything from geometry (ξ)